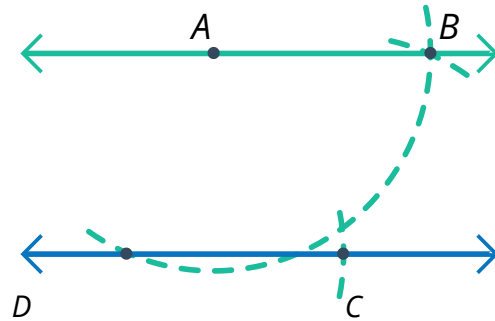


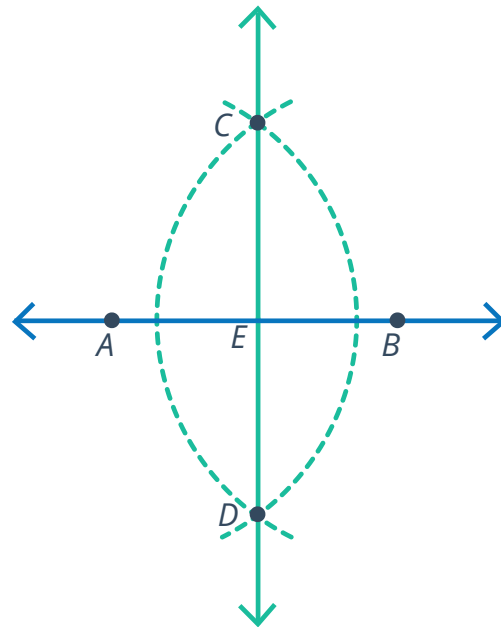
Justifying constructions using the properties of quadrilaterals

1 Consider the following diagrams. Justify the steps of the construction in a two-column proof.

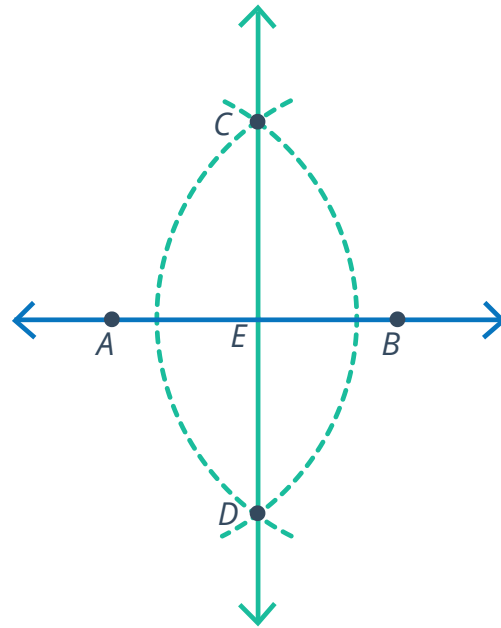
- a The line $\overleftrightarrow{AB}$ is constructed parallel to the given line $\overleftrightarrow{DC}$.



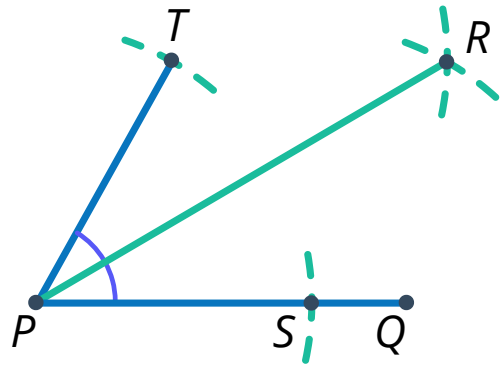
- b The line $\overleftrightarrow{CD}$ is constructed so that $\overleftrightarrow{AB} \perp \overleftrightarrow{CD}$.



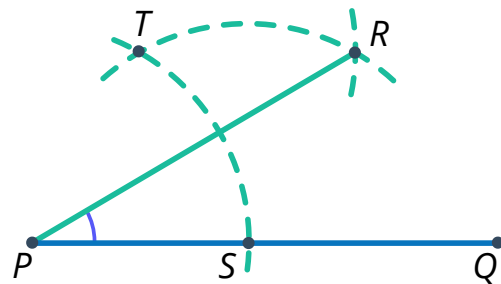
- c The line $\overleftrightarrow{CD}$ is constructed so that $\overline{AE} \cong \overline{BE}$.



- d $\overline{PR}$ is constructed so that $\angle TPR \cong \angle RPQ$.



- e $\overline{PR}$ is constructed from the given line segment $\overline{PQ}$ such that $m\angle RPS = 30^\circ$.



- 2 The inscribed square $ABCD$ is constructed from a given circle with a center at the point O . Justify the steps of the construction in a paragraph proof.

